Phase 1 · EXP-A1 — Baseline: No Ambient (Track A, ODAS-free)

experiments.pdf tag: `exp_a1_no_ambient`. **Question (Phase 1):** does ambient type drive ODAS false positives, and does matched-ambient training help? EXP-A1 is the **noise-floor baseline** — directional sources only, no ambient.

Scope of this run — Track A (ODAS-free): produced `model_a1_gt` (the GT-trained arm) entirely on this Mac: scene → render → GT clips → frozen-YAMNet head → clip accuracy. The **post-ODAS arm (`model_a1_odas`) and the FP/min metric require the `odaslive` binary (Track B)** and are deferred to a Linux/Pi env — see Status & deferred.

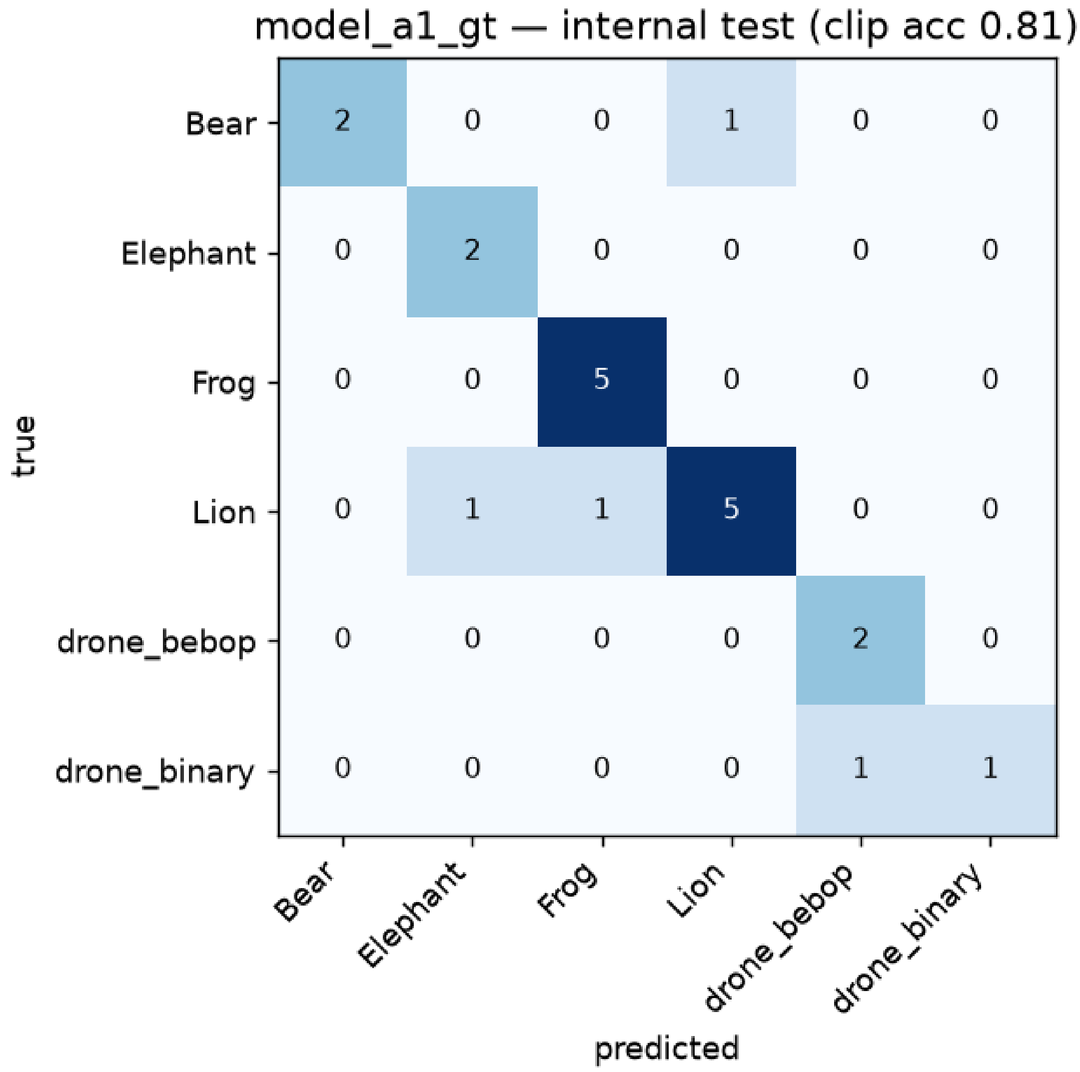
What ran

Step	Tool	Output
Scene (6 labels, 600 s, dist 10–150 m, **0 ambient**)	`expA1_render.py` (headless `renderer.py`)	90 directional sources, `.raw` + 90 `.f32` GT sidecars
GT clip dataset (3 s win, 1.5 s hop, source-level split)	`expA1_build_dataset.py` (`gt_dataset_builder.py`)	`gt_a1_no_ambient`: 133 clips, 6 labels
Holdout (300 s, seed 99)	same, `--name exp_a1_holdout`	`gt_a1_holdout`: 111 clips
`model_a1_gt` (frozen YAMNet + 256→128→softmax head)	`expA1_train_gt.py`	clip-acc + confusion

Fixed protocol honored: room 250×250×20, absorption 0.7, max_order 3, ReSpeaker 4-mic geometry, 16 kHz. Labels: Elephant, Bear, Frog, Lion, drone_bebop, drone_binary.

Result

Metric	Value
Internal test clip accuracy	**0.71–0.81** (21-clip fold, run-to-run variance)
Holdout-render clip accuracy	**0.60**
Holdout per-class F1	Elephant 0.84, Frog 0.84, drone_binary 0.48, Lion 0.45, Bear 0.38, drone_bebop 0.35



Findings

- **A clean baseline exists and the chain works end-to-end** on this Mac (render → GT dataset → trained model → eval), with real artifacts under `experiments/sim/`.
- **~20 pt generalization drop** from internal test (0.81) to a separate holdout render (0.60), *even with no ambient and clean GT audio*. The gap is driven by **Bear, drone_bebop, Lion, drone_binary**; **Elephant and Frog** hold up (≈0.84). This matches the PDF's observation that short/quiet events (Bear) are disproportionately hard, and tells us the floor is set by intra-class acoustic variability + tiny per-class counts, before ODAS even enters.
- **Drone sub-types confuse each other** (drone_binary↔drone_bebop) — expected, since both are motor/rotor spectra; per-class thresholds (Phase 3 C2) will matter here.

Caveats

- **Small dataset** (133 train / 111 holdout clips). The instance counts were modest to keep renders fast; the headline number is the *holdout* (comparable across experiments), and the internal-test figure is noisy. Scale `--instances` for tighter numbers.
- **Track A ≠ full deployment metric.** EXP-A1's headline metric in the PDF is **FP/min ≤ 2**, which is intrinsically an ODAS-detection metric (detections during quiet periods). It cannot be computed without `odaslive`. Track A gives clip accuracy only.

- **Head, not full 2-phase finetune.** This run uses the frozen-backbone Phase-1 head. The upstream `train_yamnet.py` Phase-2 (top-4 backbone unfreeze) currently breaks rebuilding the backbone from the `tensorflow/models` layer-defs (Keras-version / `_YAMNET_LAYER_DEFS` calling-convention mismatch vs current `master`). The GT-vs-ODAS comparison the experiments test is **unaffected** — the variable is the training *data*, and both arms use the identical head procedure — but the exact deployable TFLite needs the pinned `tensorflow/models` commit in the Linux env.

Status & deferred

EXP-A1 deliverable	Status
Scene + render + GT dataset	✅ done (this Mac)
`model_a1_gt` + clip accuracy	✅ done
`model_a1_odas` (post-ODAS curator dataset)	⏸ needs `odaslive` (Track B)
Event P/R + **FP/min** on holdout	⏸ needs `odaslive` (Track B)

How to reproduce

```
.venv/bin/python experiments/scripts/expA1_render.py --duration 600 --
instances 15 --seed 1 --name exp_a1_no_ambient
.venv/bin/python experiments/scripts/expA1_build_dataset.py --meta
experiments/sim/renders/renders/exp_a1_no_ambient_*.json --dataset
gt_a1_no_ambient
.venv/bin/python experiments/scripts/expA1_render.py --duration 300 --
instances 10 --seed 99 --name exp_a1_holdout
.venv/bin/python experiments/scripts/expA1_build_dataset.py --meta
experiments/sim/renders/renders/exp_a1_holdout_*.json --dataset
gt_a1_holdout
.venv/bin/python experiments/scripts/expA1_train_gt.py --dataset
experiments/sim/gt_datasets/gt_datasets/gt_a1_no_ambient --holdout
experiments/sim/gt_datasets/gt_datasets/gt_a1_holdout
```

Next

EXP-A2/A3 (synthetic Rain/Wind ambient) reuse this exact harness — just add ambient sources to the scene generator. The post-ODAS arm and FP/min for all of Phase 1 unlock once `odaslive` is built (Linux/Pi). See `08_deployment_and_loop.md` for the ODAS build.